



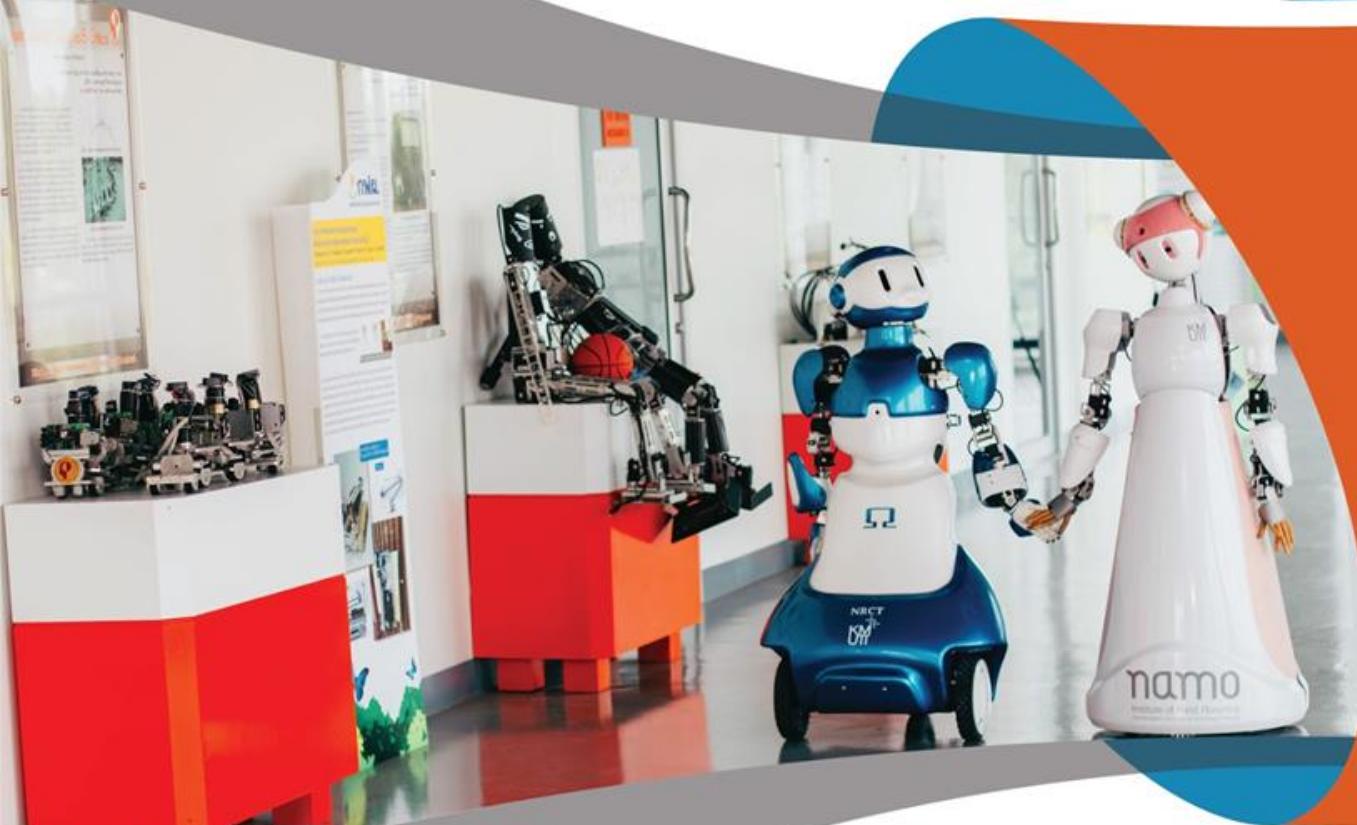
FRA532 : Mobile Robot

Lecture 5

SLAM and Navigation

Kitti Thamrongaphichartkul

Institute of Field Robotics
King Mongkut's University of Technology Thonburi
Bangkok, Thailand

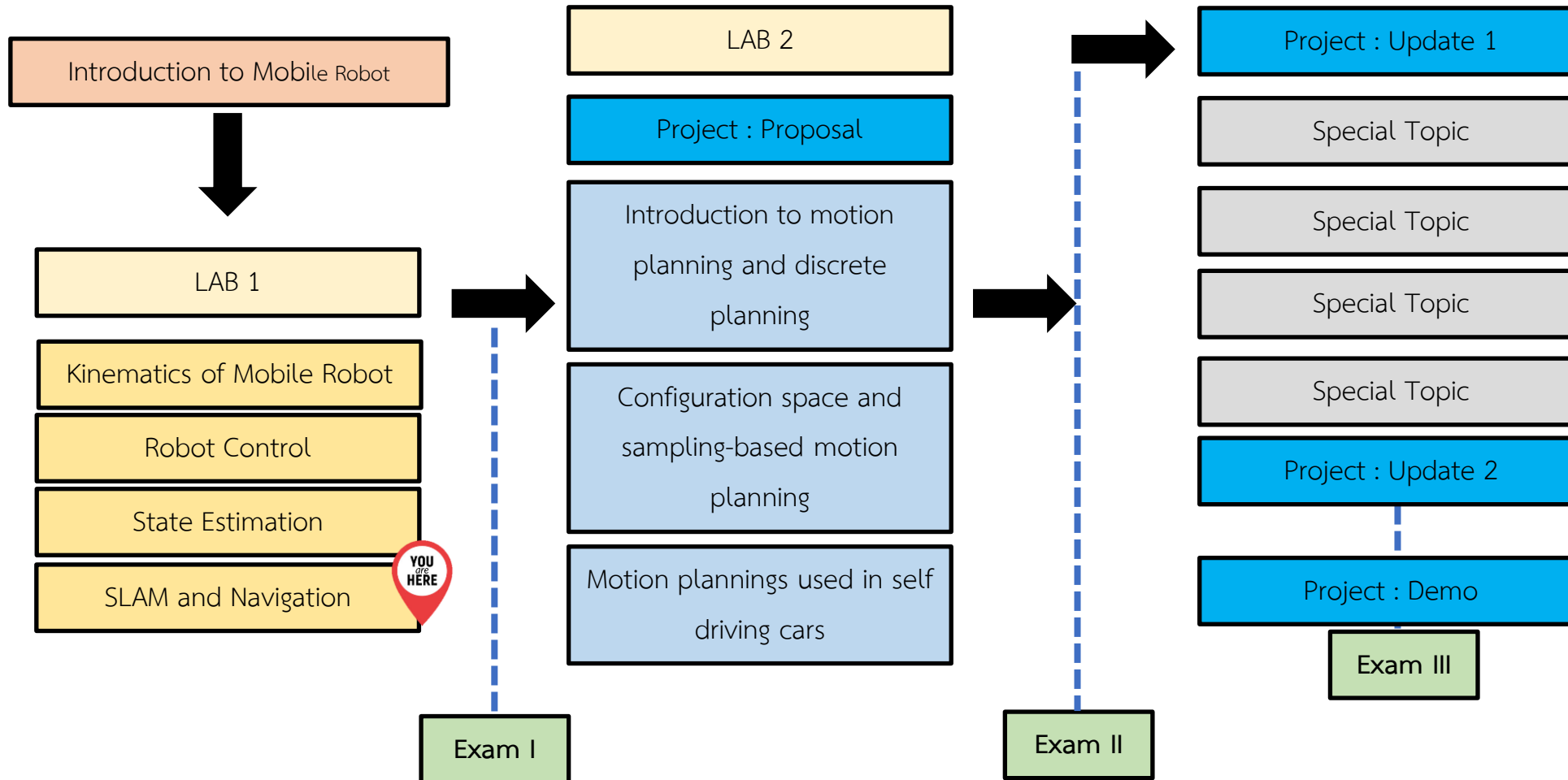


แผนการสอน

Week	Date	Lecture	Topic	Module	LAB / HW		Instructor	หมายเหตุ
					Assign	Due		
1	16-Jan-2025	1	Introduction to Mobile Robot (Motivation)				Aj.Nook	
2	23-Jan-2025	2	Kinematics of Mobile Robot		LAB 1		Aj.Nook	
3	30-Jan-2025	3	Mobile Robot Control				Aj.Nook	
4	6-Feb-2025	4	30 ปี ฟิโน				Aj.Nook	
5	13-Feb-2025	5	State Estimator				Aj.Nook	
6	20-Feb-2025		EXAM 1					
7	27-Feb-2025	6	MAP (Slam, Localization)		LAB 2	LAB 1	Aj.Nook	
8	4-March-2025	7	EXAM 1 / Hackathon Exam (24 Hour)				Aj.Nook	Project : Proposal
9	13-March-2025	8	Introduction to motion planning and discrete planning				Aj.Tee	
10	20-March-2025	9	Configuration space and sampling-based motion planning			LAB 2	Aj.Tee	
11	27-March-2025	10	Motion plannings used in self driving cars				Aj.Tee	
12	3-April-2025		EXAM 2					
13	10-April-2025	11	Project : Update 1				Aj.Nook	
14	18-April 2025	12	Special Topic I				Aj.Nook / Dummy	
15	24-April 2025	13	EXAM 2 / CBS + Nav2				Aj.Nook / Dummy	
16	1 May 2025	14	Special Topic III				Aj.Nook / Dummy	
17	8 May 2025	15	Project : Update 2				Aj.Nook	
18	15 May 2025	-	-					
19	22 May 2025	-	-					
20	29 May 2025	16	Project : Demo				Aj.Nook	

เนื้อหา

Link : <https://github.com/kittinook/MobileRobotics2025/tree/navigation>



Agenda

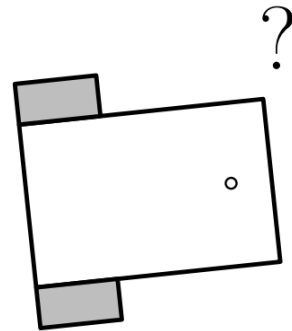
- Navigation
- Planner
- Localization
- Implementation

SLAM and Navigation

<https://github.com/kittinook/MobileRobotics2025/tree/navigation>

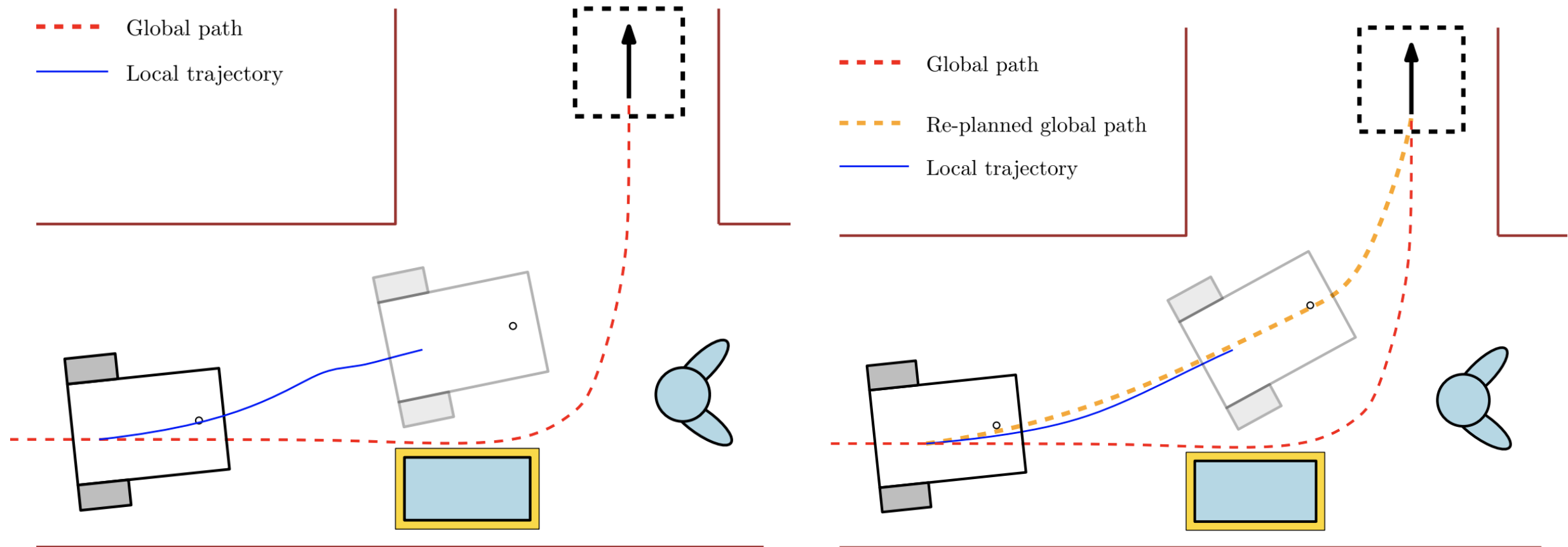
Navigation

-  Static obstacles
-  Dynamic obstacles
-  Target area

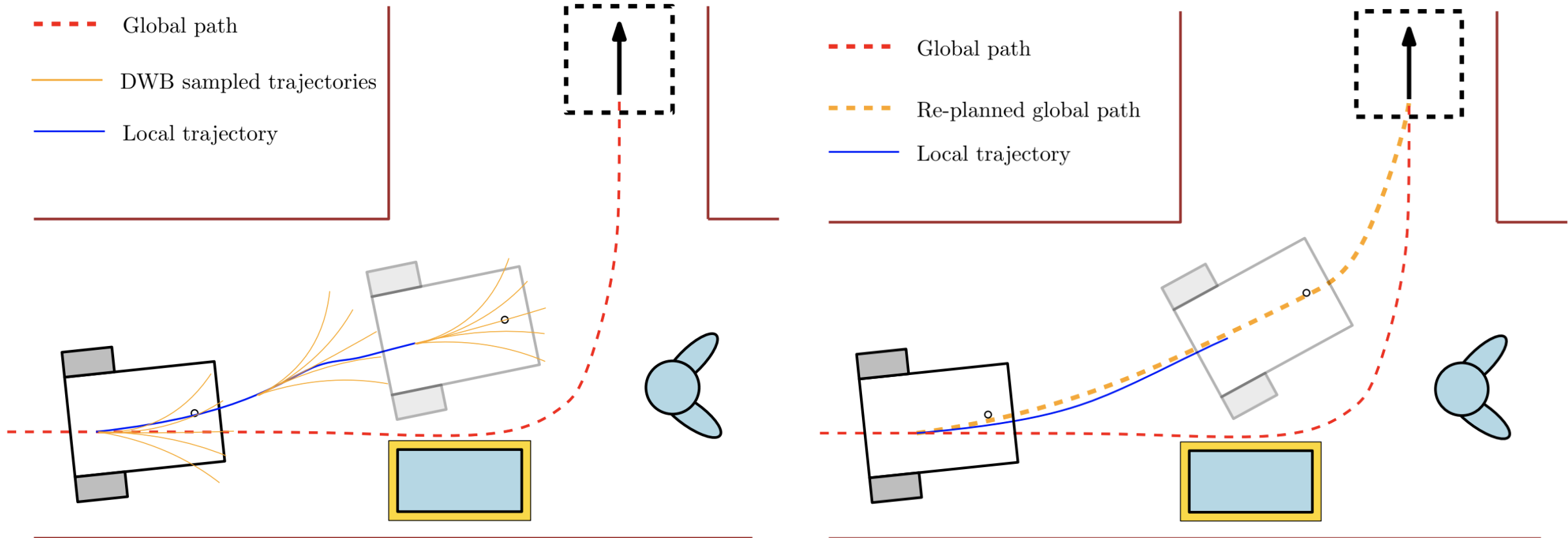


The mobile robot navigation problem: How to generate a set of velocity commands to let the robot reach its target area?

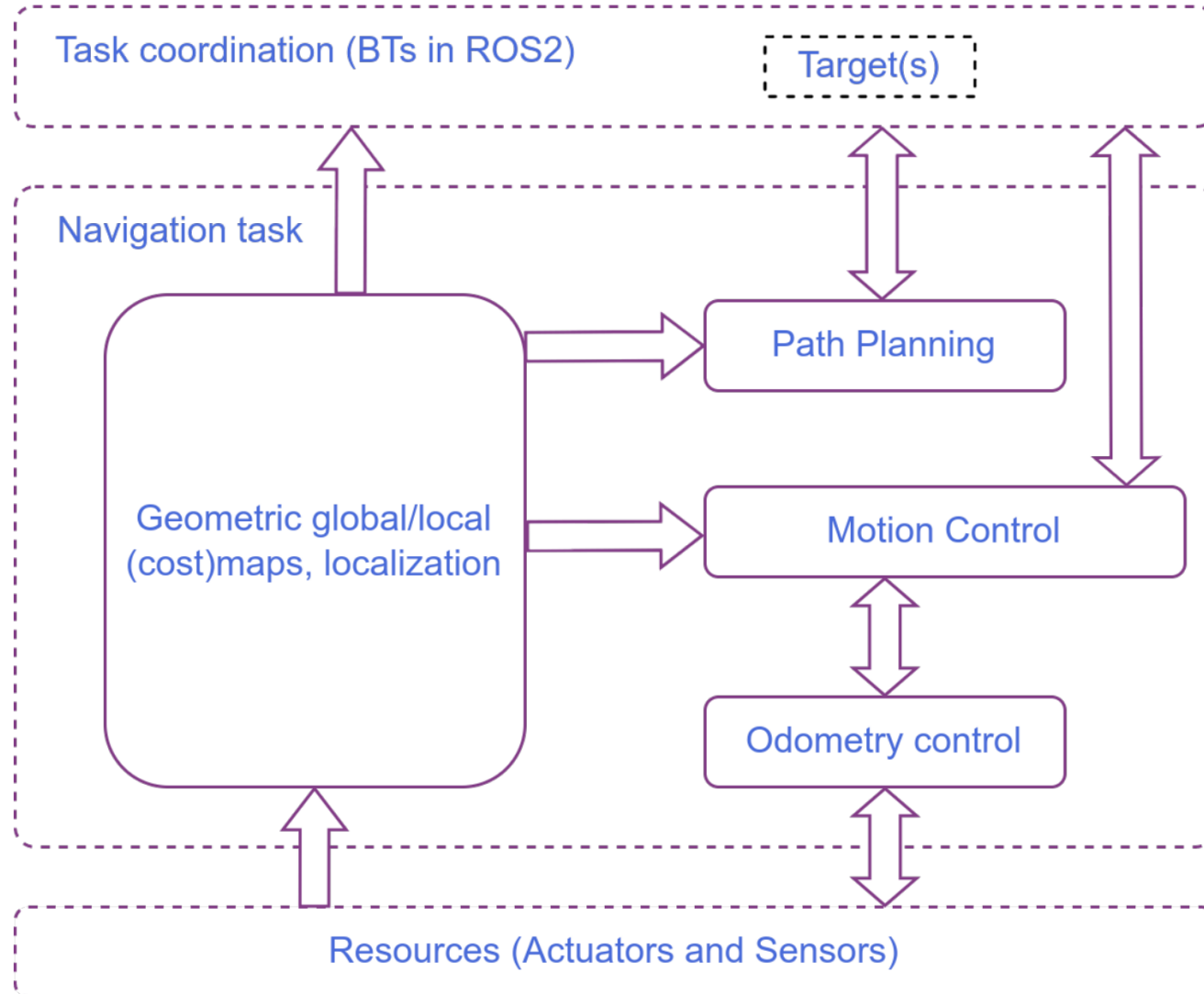
Global Planner



Local Planner

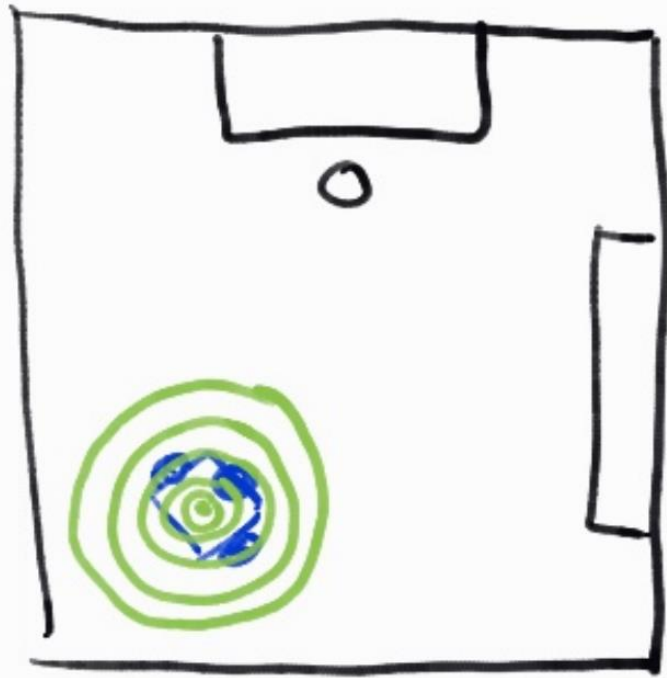


System Architecture

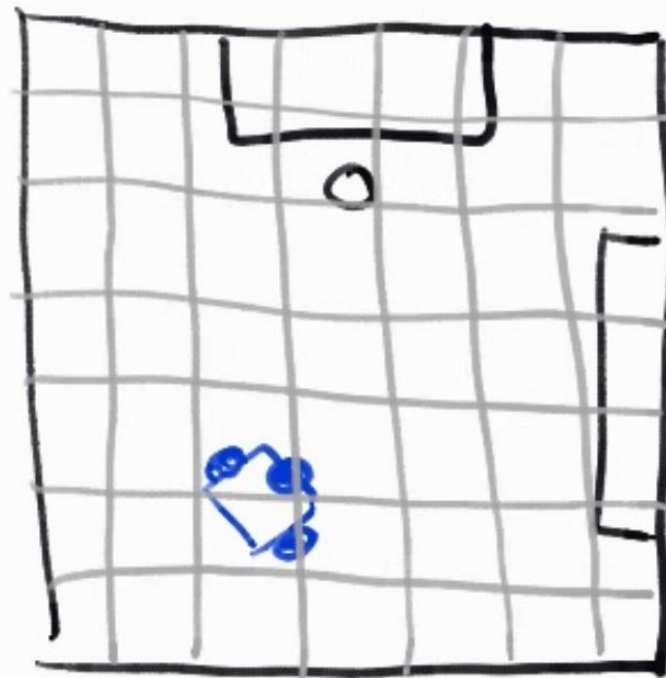


Localization

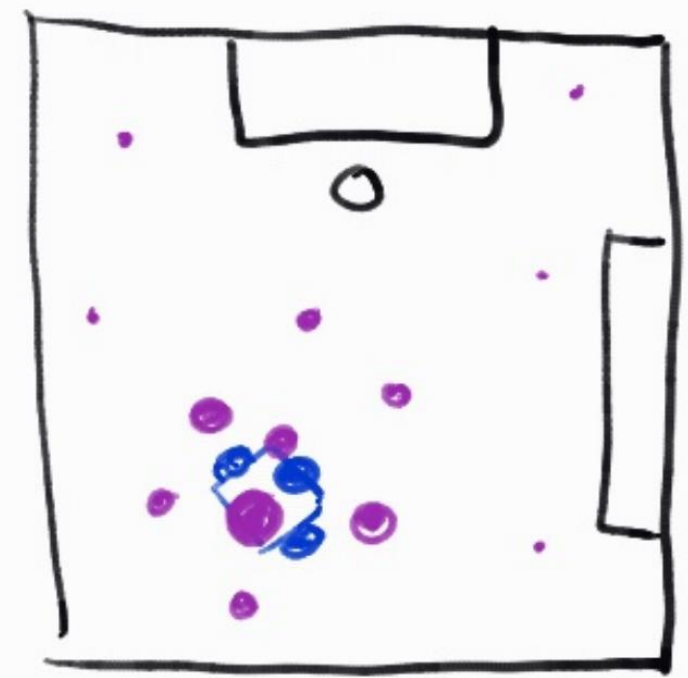
Kalman Filter



Grid based

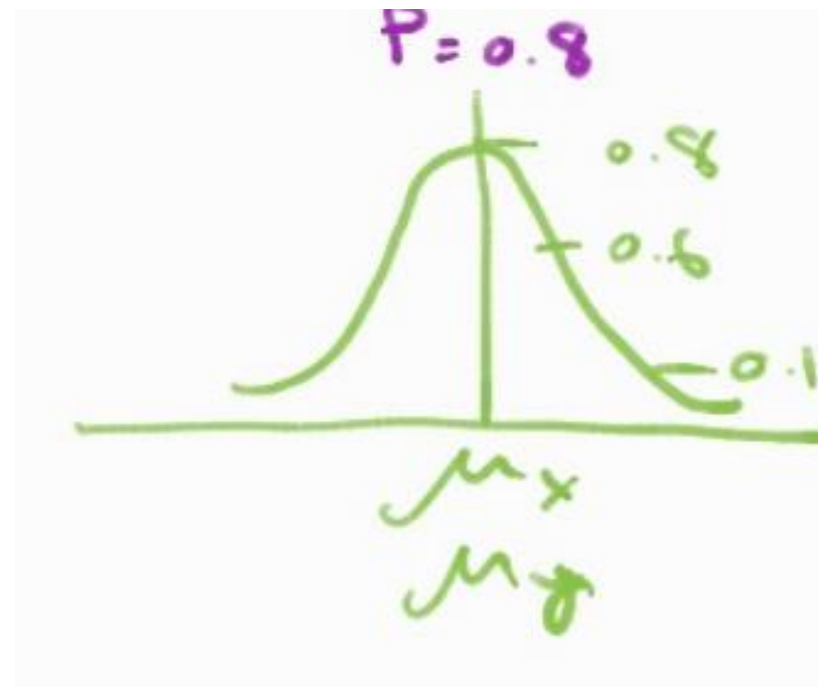


Particle Filter



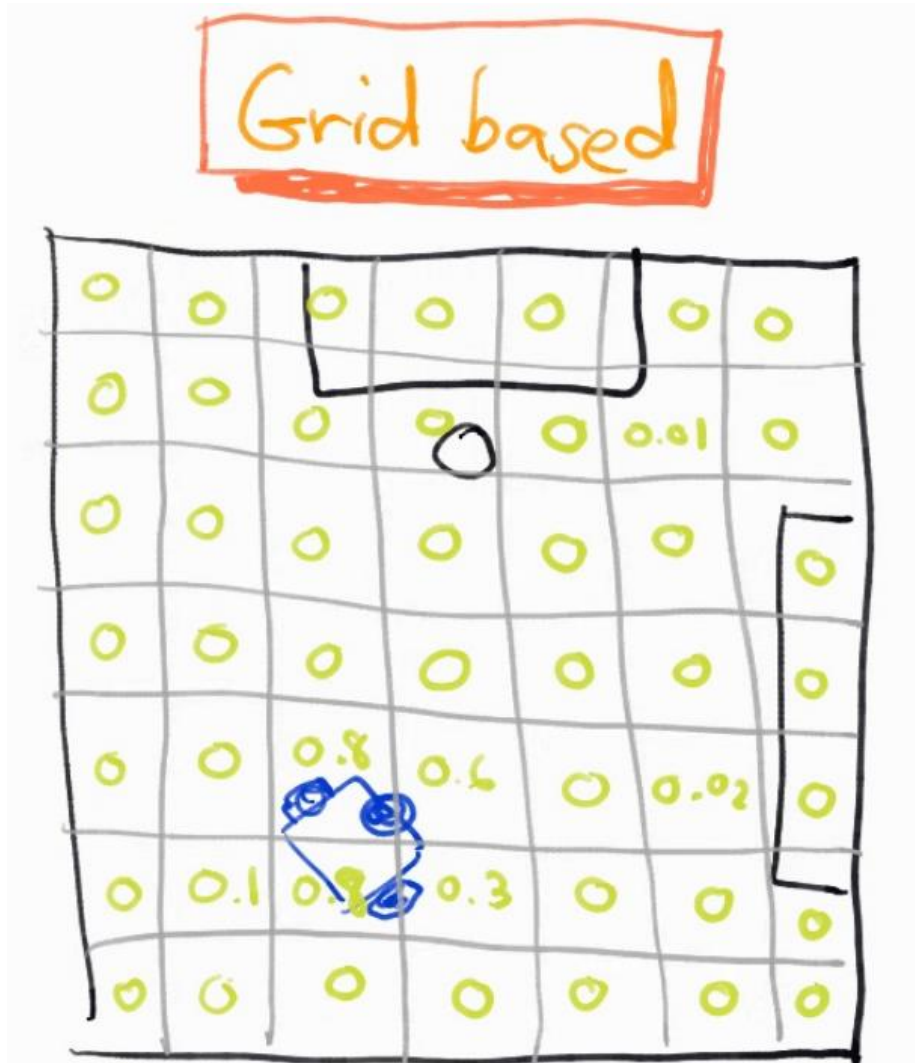
Localization

Kalman Filter



Unimodal
Gaussian

Localization



Info: $\begin{bmatrix} x \\ y \\ p \end{bmatrix}$ } state $\rightarrow$ Prob.

Localization

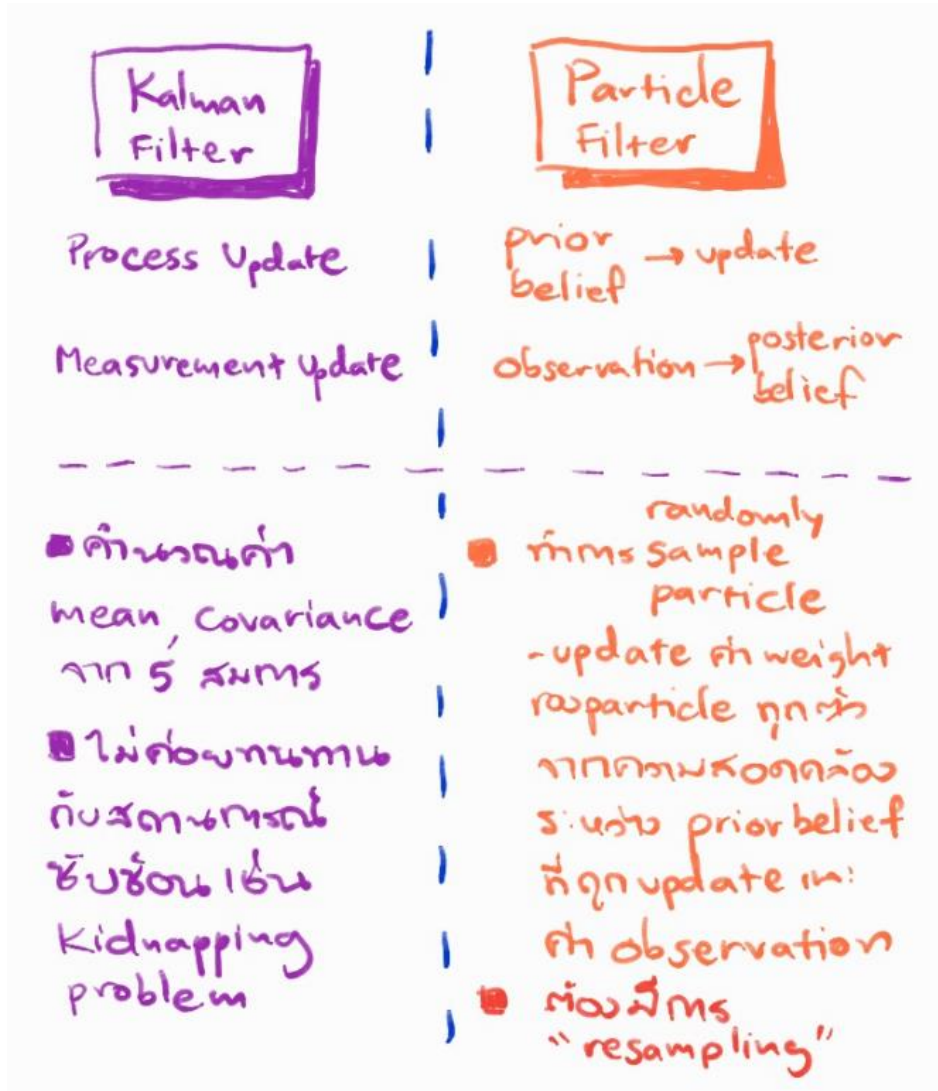
< sequential Monte Carlo estimator >

Particle Filter



Particle  $\left\{ \begin{array}{l} x = _ \\ y = _ \\ w = _ \end{array} \right\}$ state Probability "Sampling based probability distribution"

Localization



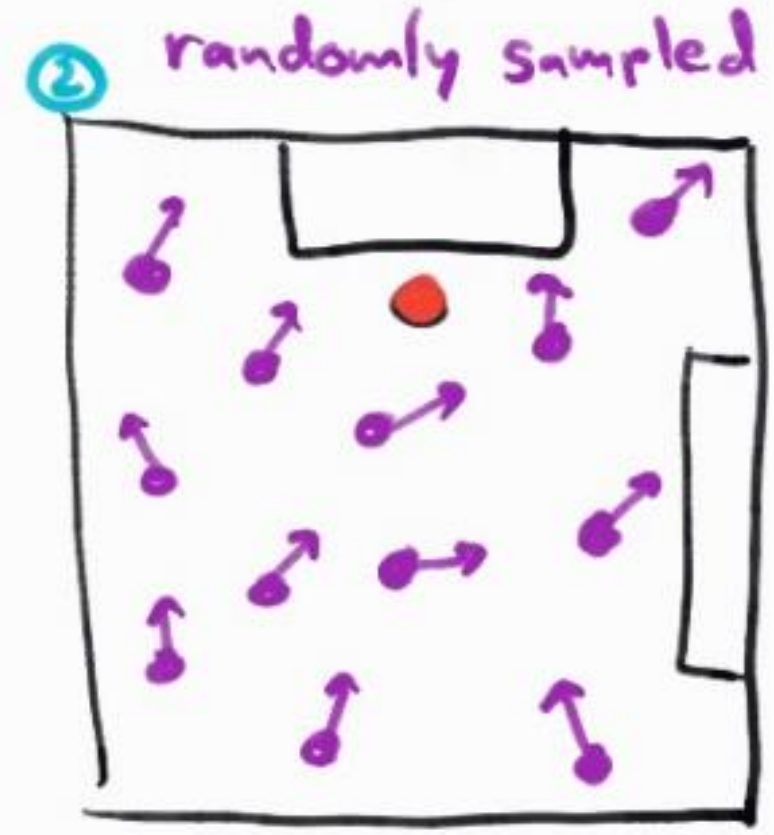
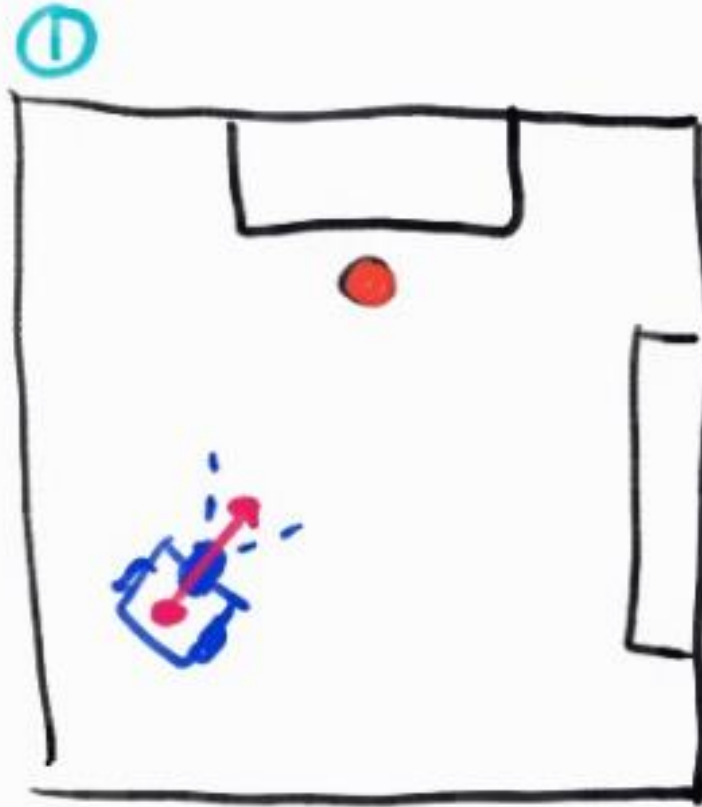
Intermittent Particle Filter

- ① δ sample $\rightarrow$ randomly generated
"N" is
- ② Importance Sampling
 - assign importance weight
"observation match in
the state & update" $\rightarrow$ weight
for each
 - normalize weight
each weight is N is total
normalize $w_j = \frac{w_j}{\sum w}$ } sum is
lot 1
- ③ Selection & Resampling
weight low $\rightarrow$ multi
resample with weight
(weight in combination, distribution)

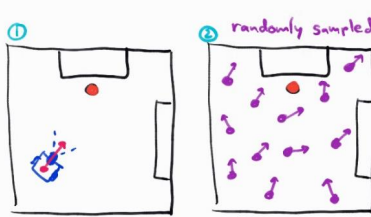
Particle Filter

ขั้นตอนการทำงานของ Particle Filter

- ① sample → randomly generated "N" particles
- ② Importance Sampling
 - assign importance weight
 - "observation match in state & update" → weight
 - normalize weight
 - normalize $w_j = \frac{w_j}{\sum w}$ } sum to 1
- ③ Selection & Resampling
 - weight low → multi
 - resample with weight (weight in binomial distribution)



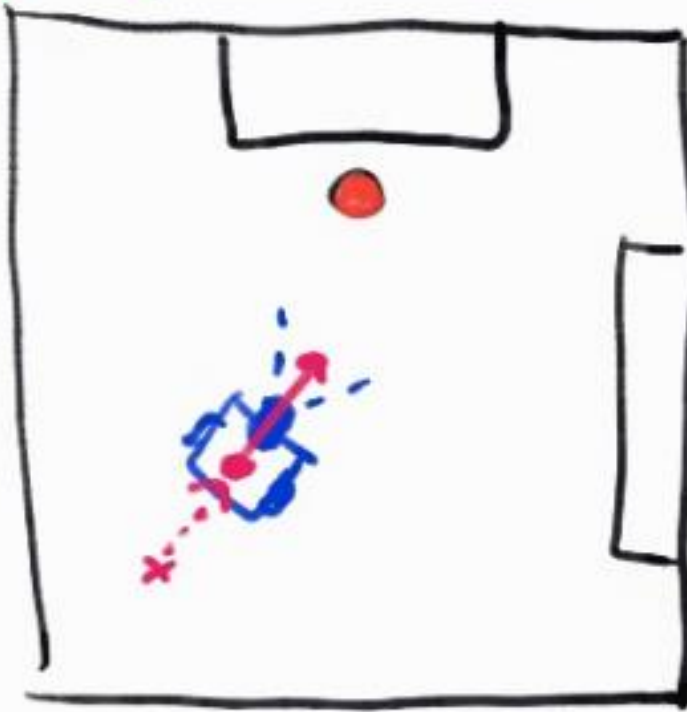
Particle Filter



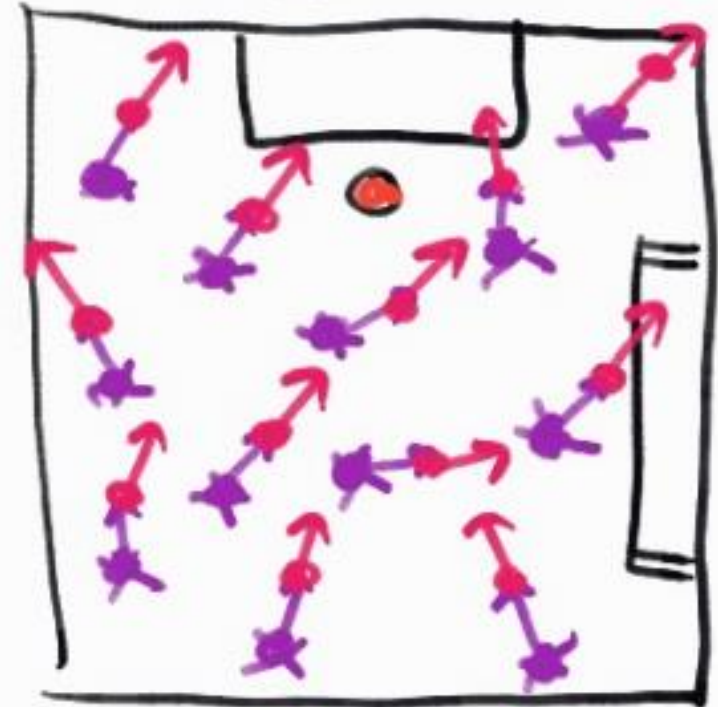
ขั้นตอนการทำงานของ Particle Filter

- ① draw sample → randomly generated "N" particles
- ② Importance Sampling
 - assign importance weight
 - "observation match to state & update" → weight
 - normalize weight
 - all weight sum N particles
 - normalize $w_j = \frac{w_j}{\sum w}$ } sum to 1
- ③ Selection & Resampling
 - weight low → multi
 - resample with new weight
 - (weight in normalized distribution)

③ select & resample particles



④ simulate to particle

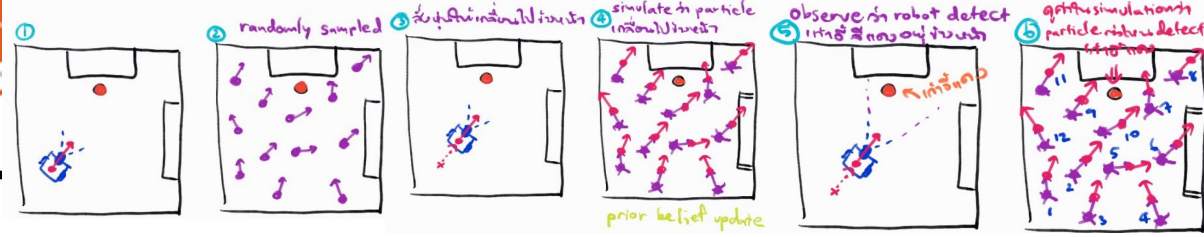


prior belief update



5. *gptf simulation with particle number detect*
- 
- The diagram shows a rectangular box representing a simulation environment. Inside the box, there is a central black dot with a red arrow pointing downwards towards it. Surrounding this central point are 12 numbered particles, each represented by a purple dot with a red arrow indicating its path. The paths are numbered 1 through 12. The paths start from the bottom and move upwards, with some paths branching or merging. The paths are numbered 1 through 12. The paths are numbered 1 through 12.

Particle Filter

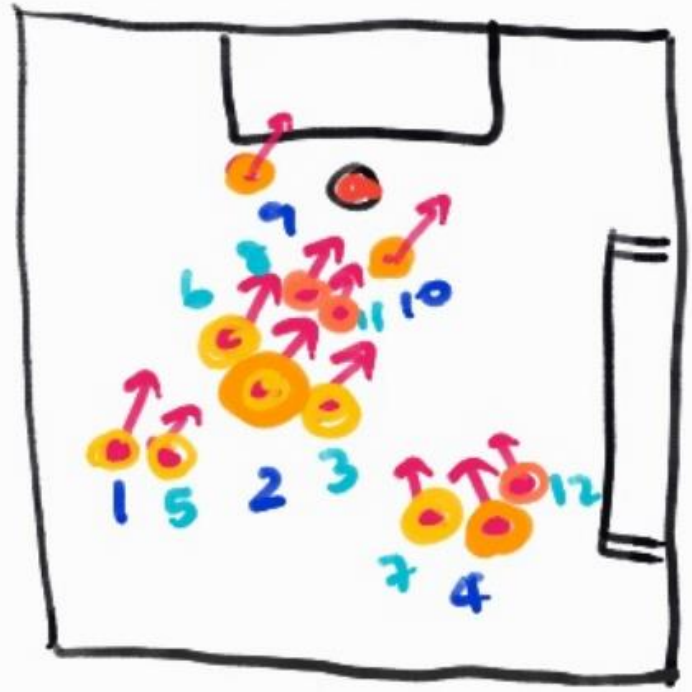
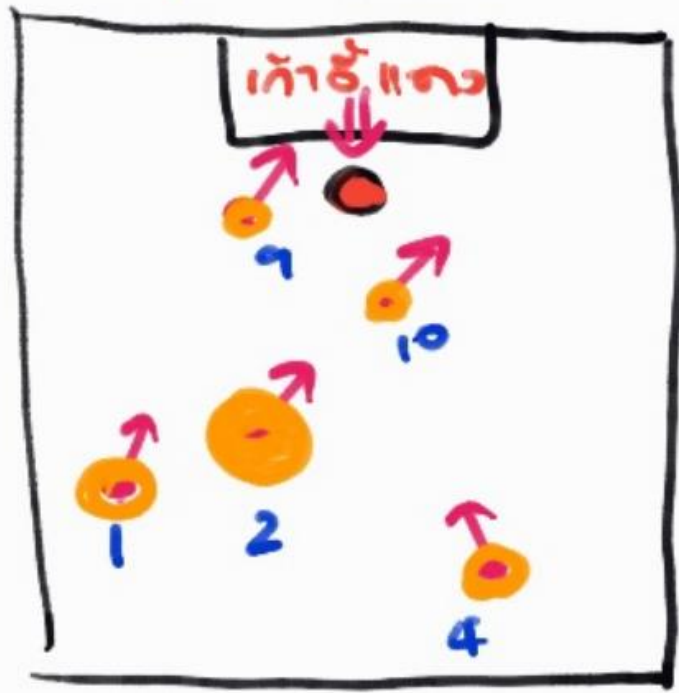


ขั้นตอนการทำงานของ Particle Filter

- ① sample → randomly generated "N" particles
- ② Importance Sampling
 - assign importance weight
 - "observation match กับ state ที่ update" → weight ที่ปรับ
 - normalize weight
 - new weight vs N particles total
 - normalize $w_j = \frac{w_j}{\sum w}$ } รวมเป็น 1
- ③ Selection & Resampling
 - weight low → multi
 - resample based on weight (weight can be normalized to, distribution)

⑦ update weight & particle & normalize

⑧ Resample new weight

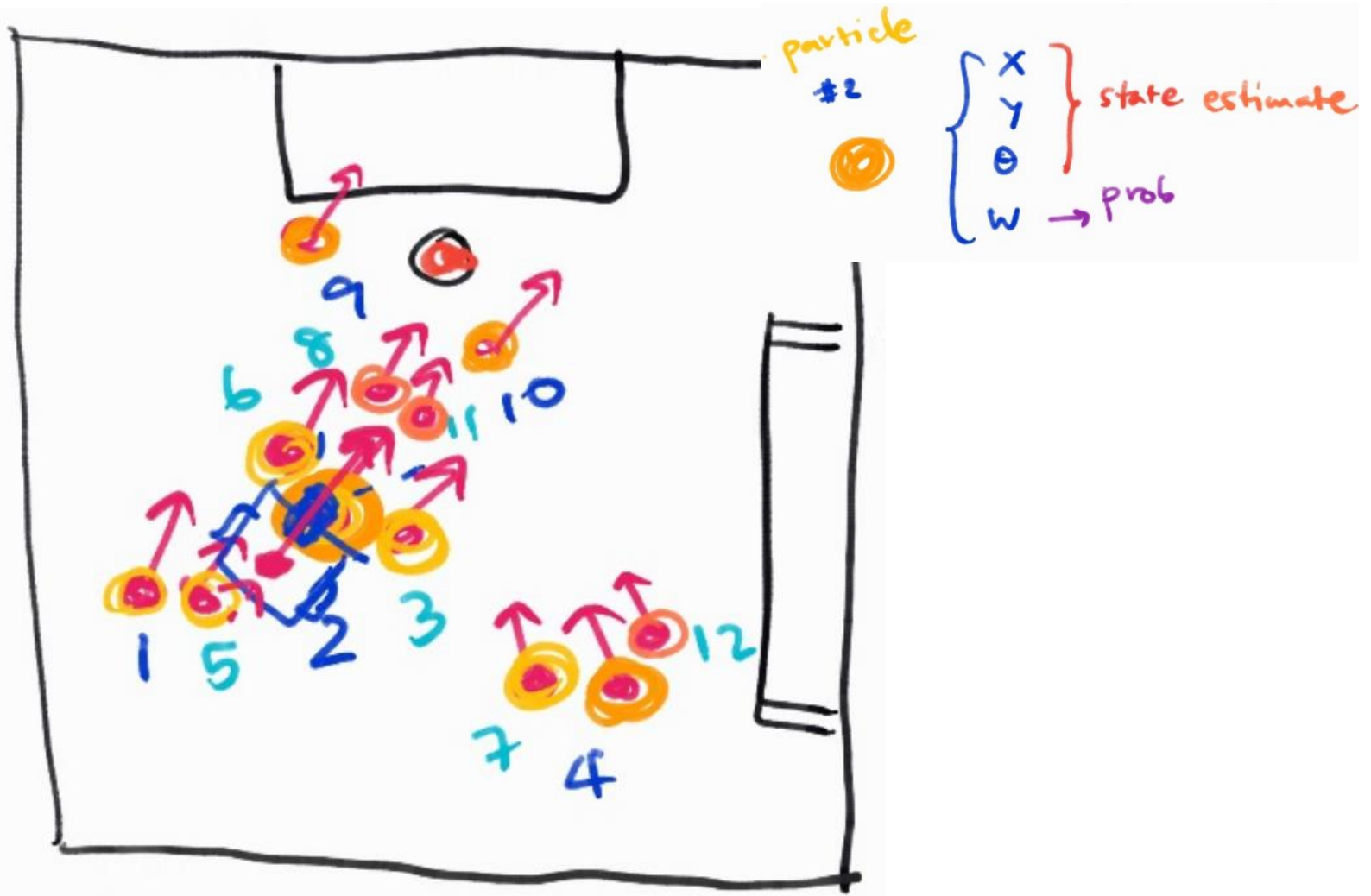


Particle Filter



Particle Filter

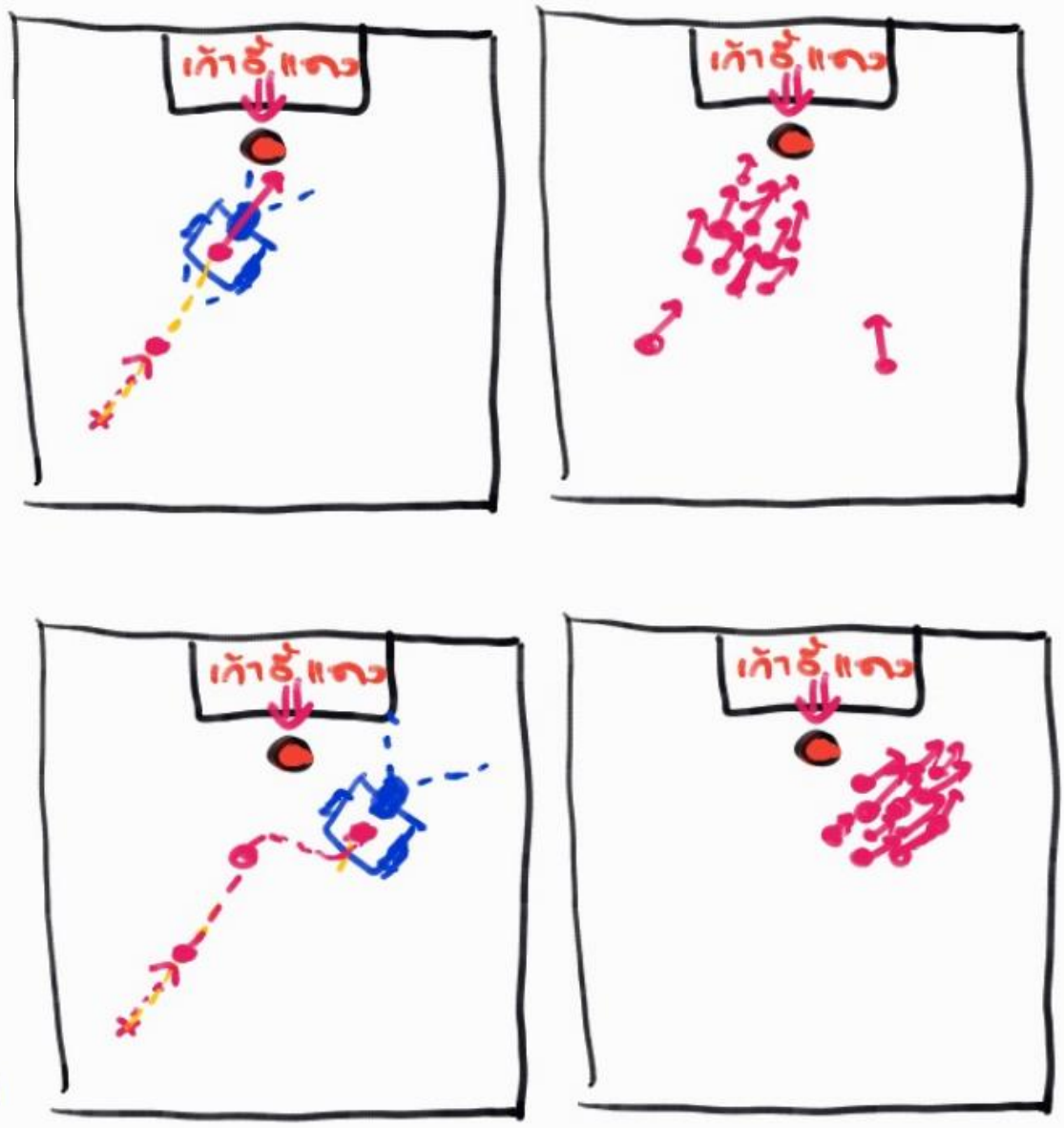
- ① sample → randomly generated "N" particles
- ② Importance Sampling
 - assign importance weight
 - "observation match in state & update" → weight
 - normalize weight
 - weight $w_j = \frac{w_j}{\sum w}$ } sum to 1
- ③ Selection & Resampling
 - weight low → multi
 - resample with weight (weight in low, high, distribution)



Particle Filter

ขั้นตอนการรัน Particle Filter

- ① สร้าง sample → randomly generated "N" particles
- ② Importance Sampling
 - คำนวณ importance weight
 - "observation match กับ state ที่ update" → weight ที่คำนวณ
 - normalize weight
 - ทุก weight รวม N particles รวม normalize $w_j = \frac{w_j}{\sum w}$ } รวมเป็น lot 1
- ③ Selection & Resampling
 - weight ต่ำ → ตัด
 - resample ใหม่ตาม weight (weight ใหม่ low/high ใหม่, กระจายใหม่)



Effective Sample Size (ESS)

$$N_{eff} = \frac{1}{\sum_{i=1}^N (w_i^i)^2}$$

ถ้า N_{eff} มีค่าน้อย → ไม่ดี

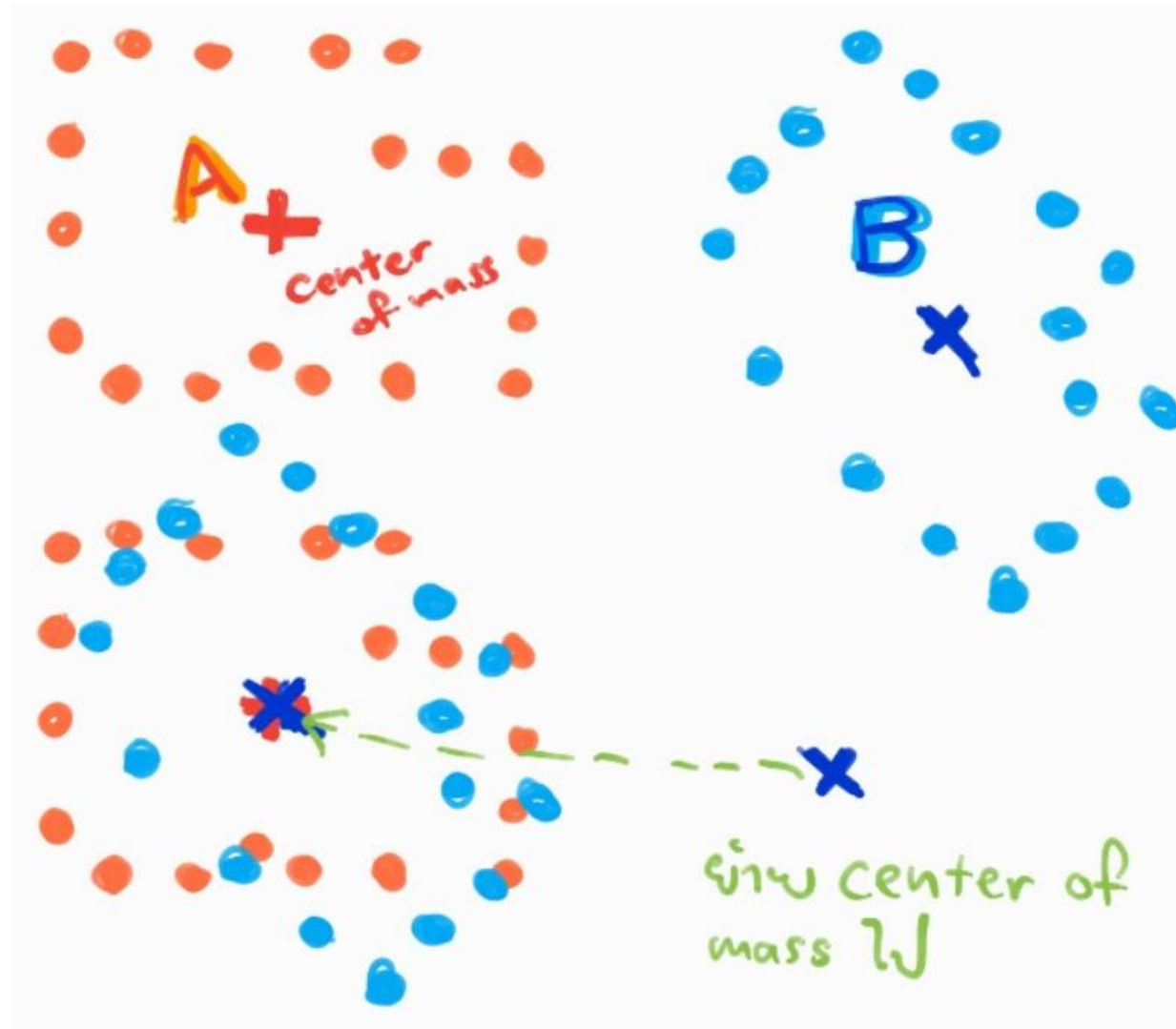
↓

"Degeneracy"

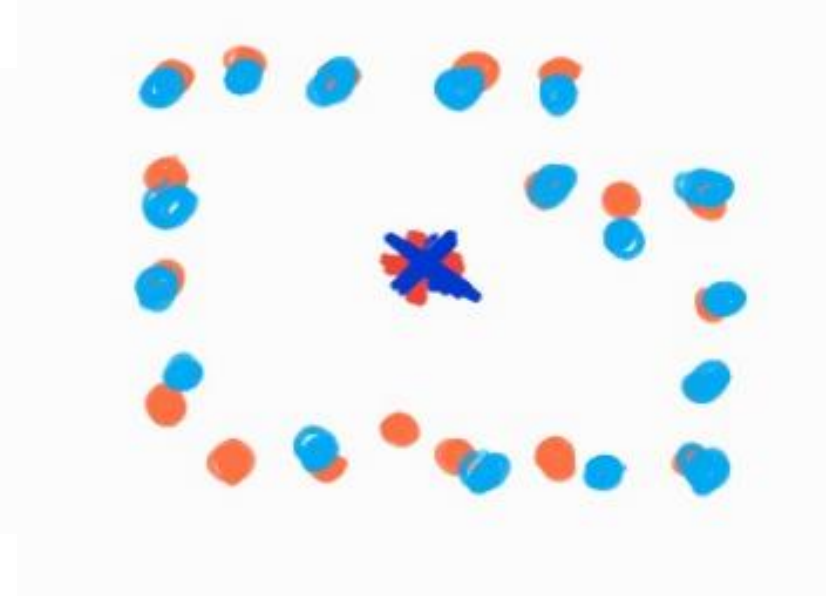
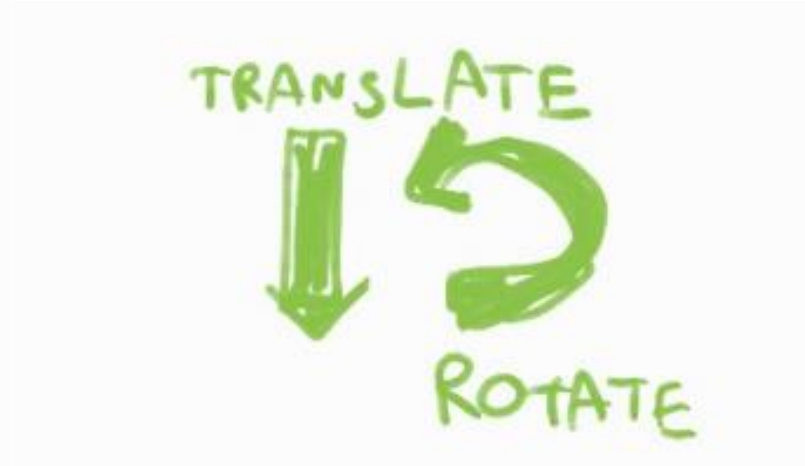
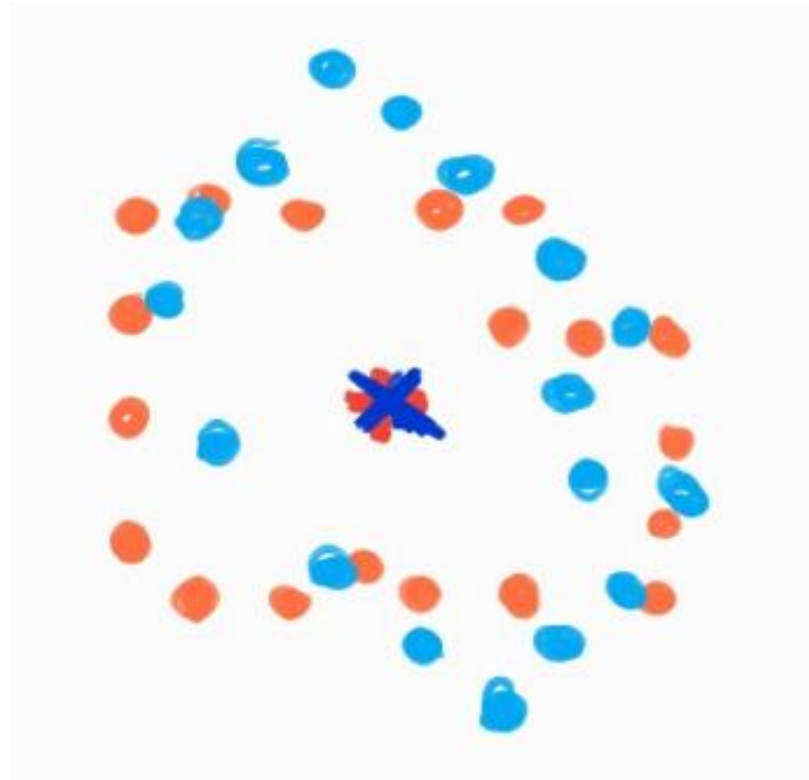
ถ้า weight มีค่า, ต่ำ → OK

0.2 0.2 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1

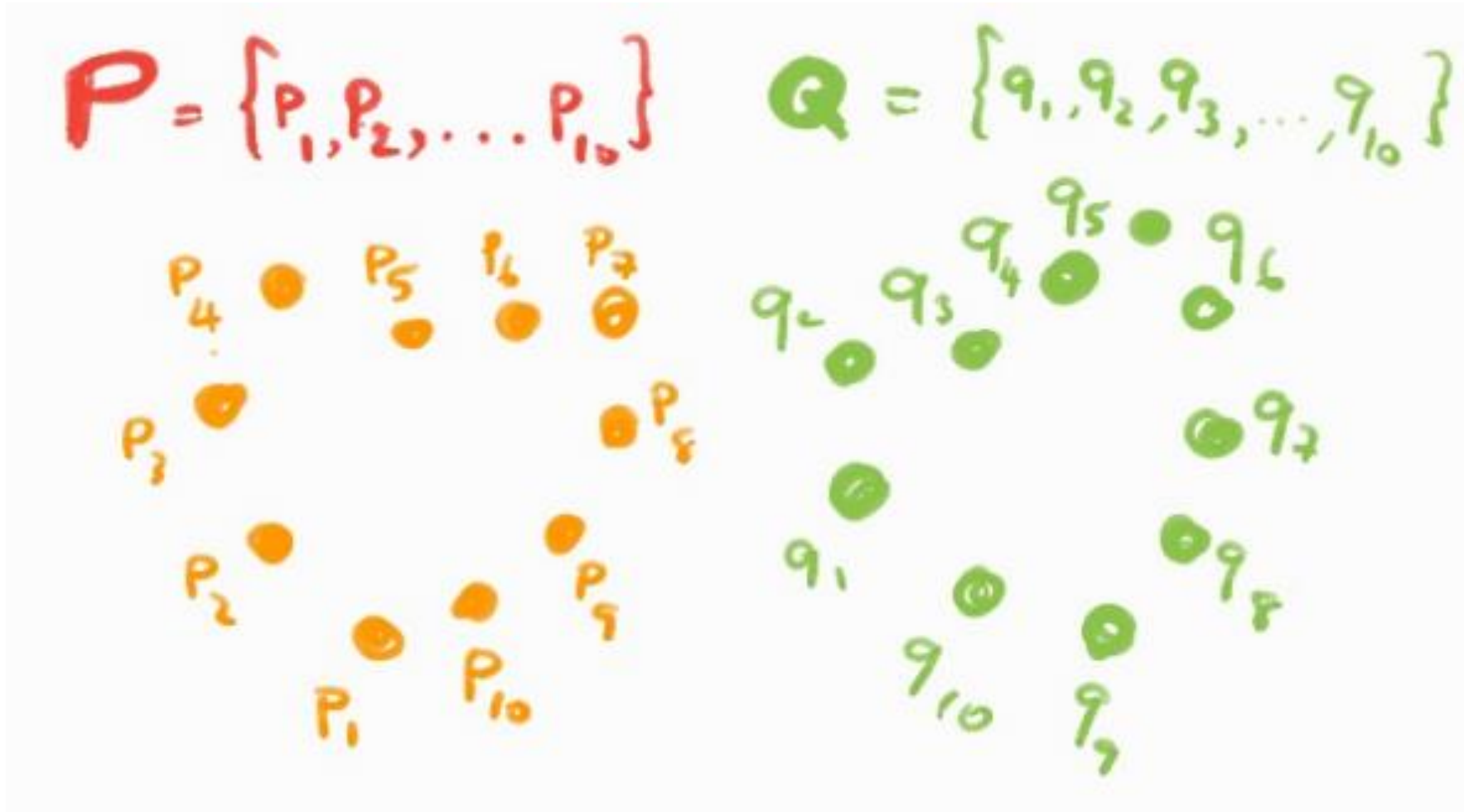
ICP (Iterative Closest Point Matching)



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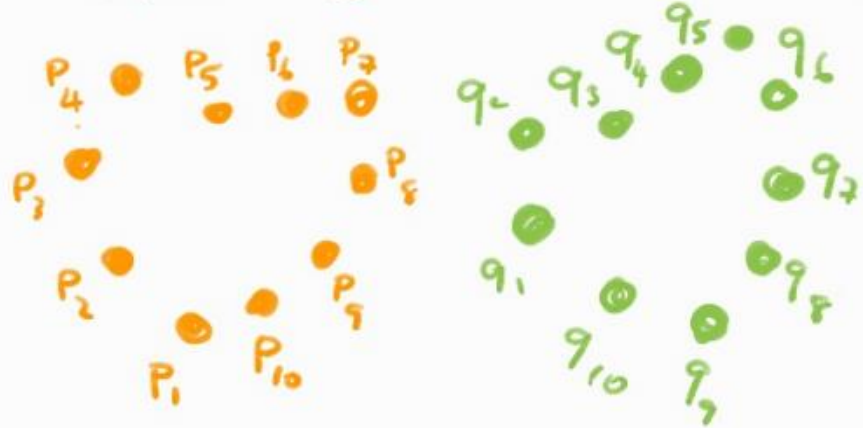


ICP (Iterative Closest Point Matching)

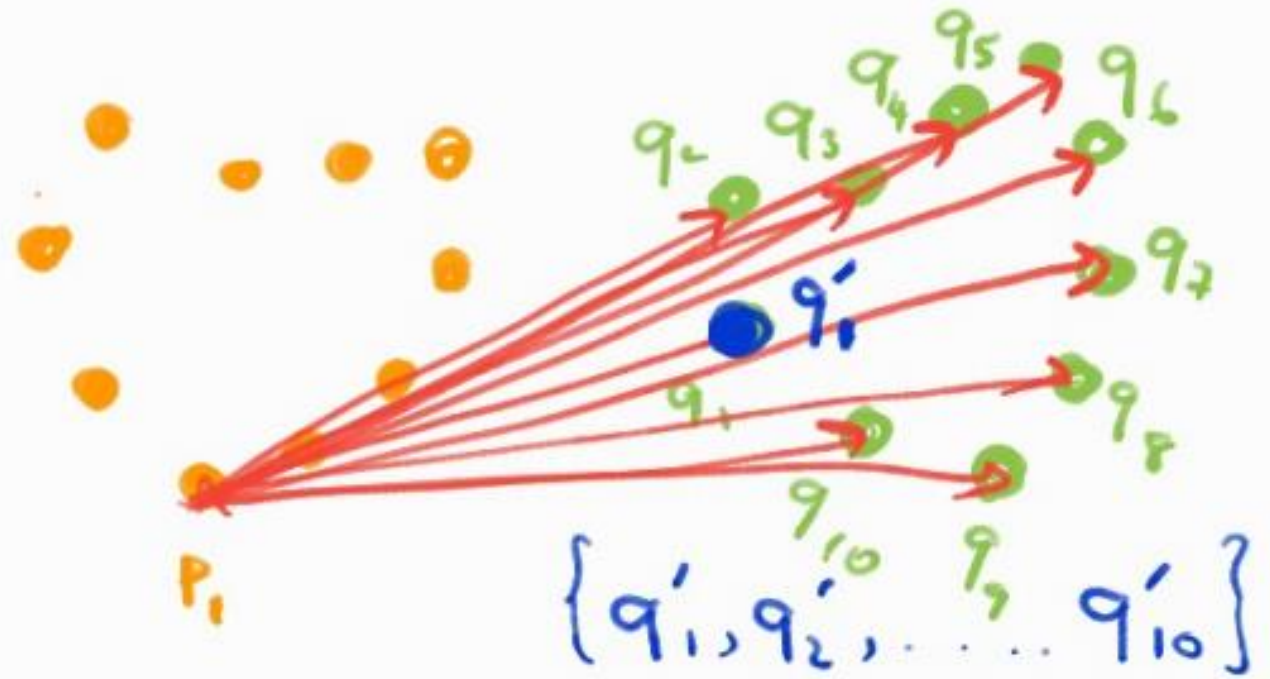


ICP (Iterative Closest Point Matching)

$$P = \{p_1, p_2, \dots, p_{10}\} \quad Q = \{q_1, q_2, q_3, \dots, q_{10}\}$$

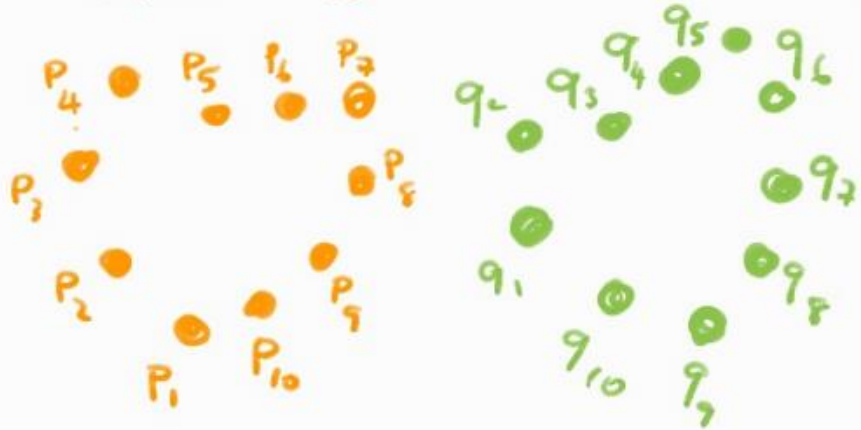


① สร้างข้อมูล Q' จากจุดใน Q ที่ใกล้ที่สุดกับจุดแต่ละจุดของ P

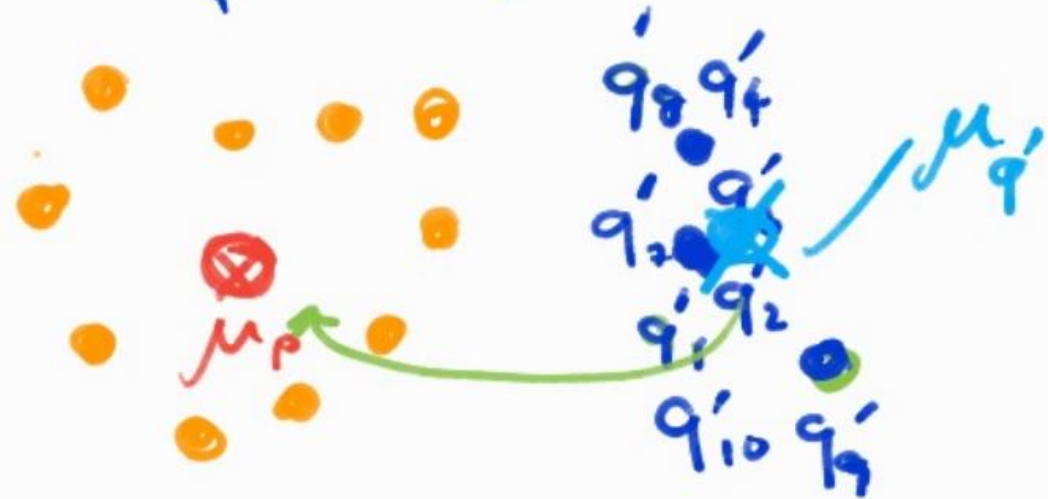


ICP (Iterative Closest Point Matching)

$$P = \{p_1, p_2, \dots, p_{10}\} \quad Q = \{q_1, q_2, q_3, \dots, q_{10}\}$$



② หา center of mass ของจุดใน P และจุดใน Q'

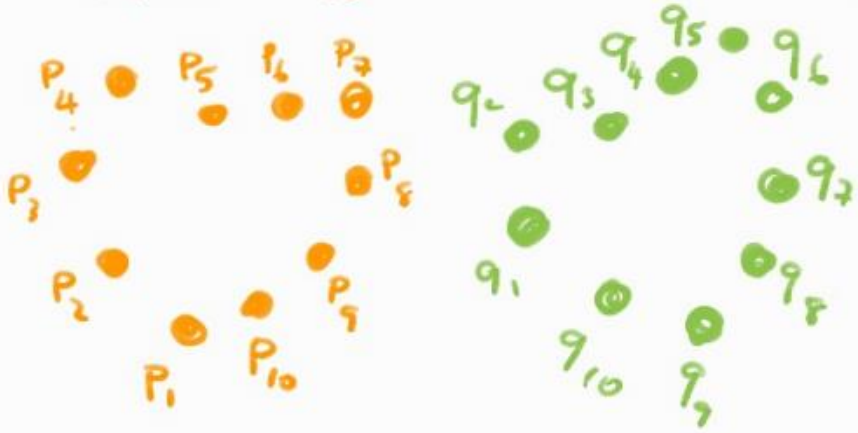


$$P' = P - \mu_P$$

$$Q'' = Q' - \mu_{Q'}$$

ICP (Iterative Closest Point Matching)

$P = \{p_1, p_2, \dots, p_{10}\}$
 $Q = \{q_1, q_2, q_3, \dots, q_{10}\}$



③ หา R และ T ใหม่

$$W = \sum_{i=1}^N \tilde{p}_i \tilde{q}_i^T$$

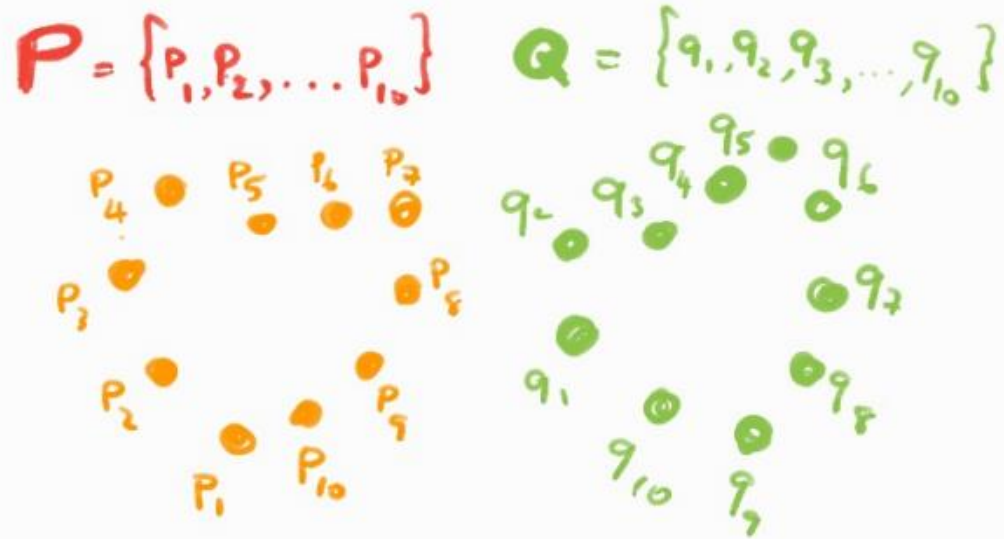
$$SVD(W) = U \Sigma V^T$$

$\begin{bmatrix} \tilde{p}_x \\ \tilde{p}_y \end{bmatrix} \begin{bmatrix} \tilde{q}_x & \tilde{q}_y \end{bmatrix}$

$$R = VU^T$$

$$T = \mu_P - R\mu_Q$$

ICP (Iterative Closest Point Matching)



④ $R_{2 \times 2} = T_{2 \times 1} P \Rightarrow \hat{P}$

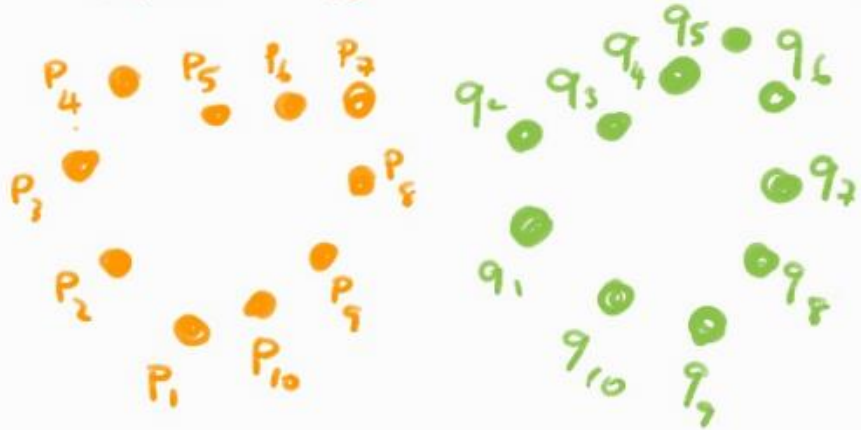
$$\begin{bmatrix} \hat{P} \\ 1 \end{bmatrix} = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} P \\ 1 \end{bmatrix}$$

หาค่า error รวม

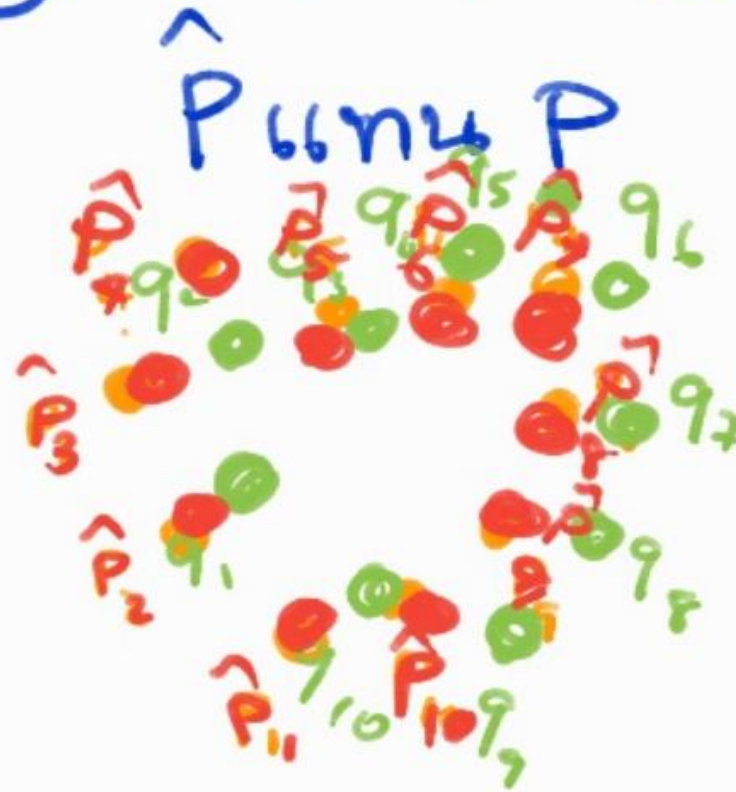
$$E = \sum_n |\hat{P} - Q'|$$

ICP (Iterative Closest Point Matching)

$$P = \{p_1, p_2, \dots, p_{10}\} \quad Q = \{q_1, q_2, q_3, \dots, q_{10}\}$$



⑤ วนทำซ้ำ ①-④ โดยซ้ำ





Q & A

A Cradle of Future Leaders in Robotics